

Element K

Reflection on the design project

The project designer provides a consistently clear, insightful, and comprehensive reflection on, and value judgment of, each major step in the project; the reflection includes a substantive summary of lessons learned that would be clearly useful to others attempting the same or similar project.

We presented our final design at Brentwood Highschool to the zoo staff. During project we presented the problem, our assessment of the problem, planning and scheduling, our decision matrix, as well as all 3 iterations of our projects. We planned our work strategy to ensure we had a clear path forward. Our initial work plan laid out the steps we intended to follow; however, as we progressed, we found it necessary to adapt and modify several aspects of our original plan. Despite these changes, we successfully developed a prototype and are now advancing towards the creation of the final product.

The Planning and Scheduling

- Before we even began to even decide on what path to take, we decided to plan out relatively what our work plan should look like.
- Unfortunately, we had to change multiple aspects of our initial plan, but upto-date, we were able to finalize a prototype and a starting to work on the final product.

A hand-drawn Gantt chart on a whiteboard. The chart has a grid with 7 rows and 8 columns. The columns are labeled: Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, and Sunday. The rows are numbered 1 through 7. The tasks and their durations are as follows:

Task	Start	End
Research	Monday	Monday
Design	Tuesday	Tuesday
Prototype	Wednesday	Wednesday
Final Product	Thursday	Thursday
Testing	Friday	Friday
Final Product	Saturday	Saturday
Final Product	Sunday	Sunday

The Nashville Zoo, specifically the Andean Bear Keepers, presented us with a set of requirements for the design of an enrichment device for the Andean Bears. The primary goals were to create a device that would enrich and entertain the bears within their exhibit, could be attached to the enclosure via a chain, included a reward factor, and

offered diverse uses so it could be employed multiple times throughout the bears' stay.

The Problem

- The Nashville Zoo, specifically the Andean Bear Keepers, requested for the following requirements for the design of an enrichment device for the Andean Bears:
 - Entertain or enrich the experience of the bear within the exhibit
 - The ability to be attached to the enclosure/ exhibit via a chain
 - Contain somewhat of a reward factor
 - Diversity in which the device can be used multiple times throughout the bear's stay.



To tackle these requirements effectively, one of our initial objectives was to utilize a decision matrix. This tool proved invaluable, allowing us to gather feedback and optimize our efforts both in the classroom and during extracurricular hours. The decision matrix helped us evaluate various design options systematically and choose the best path forward based on a set of criteria that aligned with the project goals.

Decision Matrix

- One of the first goals we set for ourselves was to actually use a decision matrix.
- This allowed us to use feedback and best optimize our time within the classroom and sometimes outside of class.

Criteria	Plan orb	Omni Orb	Trunk Cam	Periw
Size	1	1	2	1
Time to solve	4	3	3	5
Weight	3	4	3	1
Durability	3	4	3	2
Availability	4	4	3	3
Cost	3	3	4	2
Assembly Time	2	2	3	5
Difficulty to build	2	2	4	2

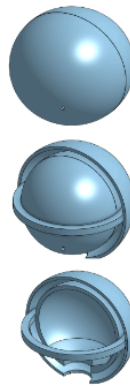
Key: * Each criteria is 1-5

Our first design iteration, named "The Omniwheel," aimed to engage the bears by teaching them to align holes in the device so that food would fall out. The movement of the device was controlled by an omni-wheel acting as a gyroscope, with an inner circle weighted at the bottom to ensure it always pointed downwards. Despite the innovative

concept, we realized that this design was overly complex and not as feasible for the bears'

Iteration I: Omni-wheel : Initial Design

- The first iteration of our solution
- Main Idea was to teach the bears to align the holes so the food would fall out
- Movement controlled by omni-wheel acting as a gyroscope
- Inner circle is weighted at the bottom so it is always pointing down

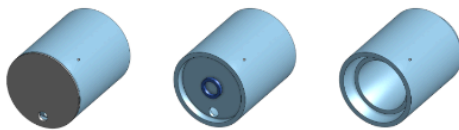
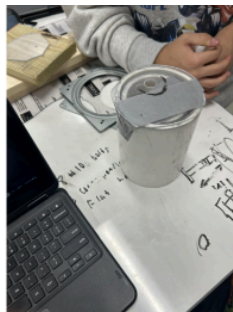


use.

Learning from the first iteration, we moved to the second design, "The Twin Drum." This iteration retained the goal of rewarding the bears with food but simplified the mechanics, making the device less complex and more cost-effective. In this design, the bears would align two drums to release food, with movement controlled by a large ball bearing and a lever that the bears could grab or bite. This iteration was more practical but still had room for improvement.

Iteration II: Twin-drum : Simplification

- The second iteration of our solution
- Same Goal as original design but far less complex and expensive
- Bears now align the drums to get food to come out
- Movement now controlled by a large ball bearing and a lever which the bears grab or bite onto



Our third and final iteration, "The Spin Drum," further refined the concept by eliminating the unnecessary central drum and focusing on the “doorway” mechanism. This design reinforced the drum internally with metal poles and covered the bevels with a sheet to prevent food from getting trapped. This final iteration addressed the complexity and durability issues identified in the earlier versions, resulting in a more robust and user-friendly enrichment device for the bears.

Iteration III: Spin-drum : Final Design

- The third and final iteration of our solution
- Removed the unnecessary central drum and is now fully based around the “doorway”
- Drum will be reinforced on the inside by metal poles and bevels will be filled by a sheet to keep food from falling in them

